

3.5 Block Matrix Multiplication



Core skills you'll master in this section:

- ▶ Block matrix multiplication: the shapes of the sub-matrices must match.
- ▶ The first perspective on matrix multiplication: the inner product.
- ▶ Extending the first perspective: split A into several rows, split B into several columns.
- ▶ The second perspective on matrix multiplication: the outer product.
- ▶ Extending the second perspective: split A into several columns, split B into several rows.
- ▶ The third perspective on matrix multiplication: a linear combination of column vectors.
- ▶ The fourth perspective on matrix multiplication: a linear combination of row vectors.

The four perspectives on matrix multiplication introduced earlier are, in essence, all different forms of block matrix multiplication. Building on a summary of these first four forms of block matrix multiplication, this section further extends the discussion to other commonly used forms of block matrix multiplication.

Table 1 compares the four perspectives on matrix multiplication – please compare and review them on your own.

Table 1. Comparing the Four Perspectives on Matrix Multiplication

$AB = C$	First Perspective	Second Perspective	Third Perspective	Fourth Perspective
Description	Inner product	Outer product	Column-vector linear combination	Row-vector linear combination
Partition	<i>A sliced into rows</i> <i>B sliced into columns</i>	<i>A sliced into columns</i> <i>B sliced into rows</i>	<i>A sliced into columns</i>	<i>B sliced into rows</i>
Mathematical expression	$c_{i,j} = a^{(i)}b_j$	$C_k = a_k b^{(k)}$	$c_j = Ab_j$	$c^{(i)} = a^{(i)}B$
Meaning	<i>The inner product of A's row i and B's column j</i>	<i>The sum of the outer-product matrices of A's column k and B's row k</i>	<i>A's column vectors combined with weights from each column of B</i>	<i>B's row vectors combined with weights from each row of A</i>
Shape	Determined by A's row count $\times$ B's column count	Each outer-product result is $m \times n$, matching the final matrix	<i>A's column vectors have the same dimension as C's column vectors</i>	<i>B's row vectors have the same dimension as C's row vectors</i>

The First Perspective on the Matrix Multiplication AB : A Sliced into Rows, B Sliced into Columns

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Comments and corrections are welcome. The book's dedicated email address: jiang.visualize.ml@gmail.com

For the matrix multiplication $\mathbf{AB} = \mathbf{C}$, slicing the left-hand matrix $\mathbf{A}$ into rows and the right-hand matrix $\mathbf{B}$ into columns — as shown in Figure 1, we're actually writing the matrix multiplication $\mathbf{AB}$ as $m \times n$ separate matrix multiplications.

Each such matrix multiplication, $\mathbf{a}^{(i)} @ \mathbf{b}_j = c_{ij}$ yields one element of the result $\mathbf{C}$.

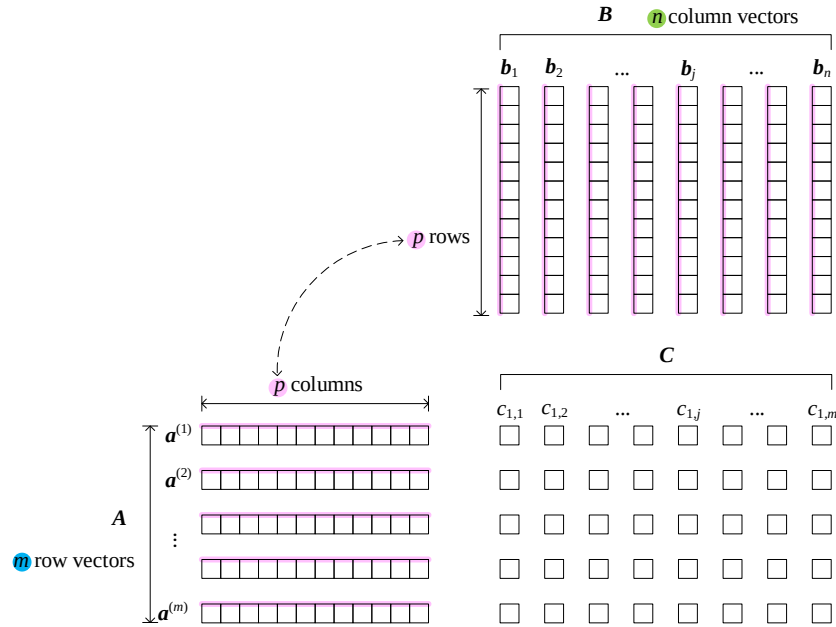


Figure 1. The Matrix Product $\mathbf{AB}$. $\mathbf{A}$ Sliced into Rows, $\mathbf{B}$ Sliced into Columns

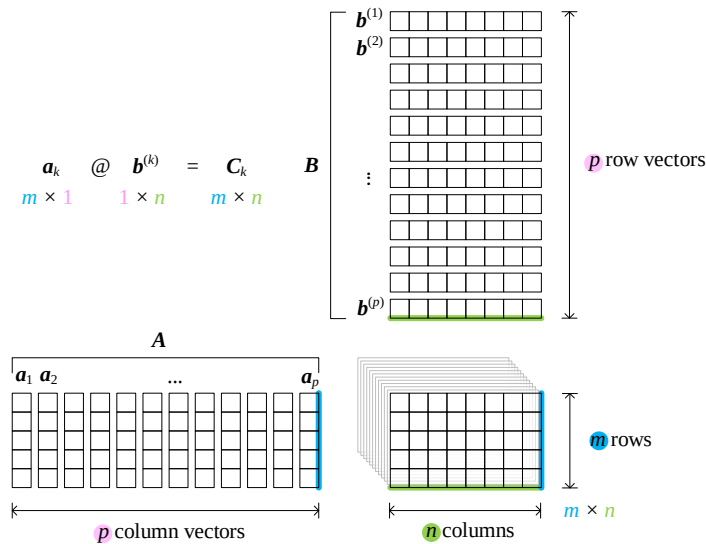
The Second Perspective on the Matrix Multiplication $\mathbf{AB}$: $\mathbf{A}$ Sliced into Columns, $\mathbf{B}$ Sliced into Rows

For the matrix multiplication $\mathbf{AB} = \mathbf{C}$, slicing the left-hand matrix $\mathbf{A}$ into columns and the right-hand matrix $\mathbf{B}$ into rows — as shown in Figure 2, we're actually writing the matrix multiplication $\mathbf{AB}$ as a superposition of p separate matrix multiplications.

Each such matrix multiplication, $\mathbf{a}_k @ \mathbf{b}^{(k)} = \mathbf{C}_k$ yields a matrix with exactly the same shape as the result $\mathbf{C}$.

Summing the p $\mathbf{C}_k$ together gives matrix $\mathbf{C}$.

p corresponds to the number of columns of matrix $\mathbf{A}$, which is also the number of rows of matrix $\mathbf{B}$ — this is the dimension that gets “canceled out” in the matrix multiplication $\mathbf{AB}$.

Figure 2. The Matrix Product AB . A Sliced into Columns, B Sliced into Rows

The Third Perspective on the Matrix Multiplication AB : B Sliced into Column Vectors

For the matrix multiplication $AB = C$, slicing the right-hand matrix B into columns – as shown in Figure 3, every column of the matrix-multiplication result C is a linear combination of A 's column vectors.

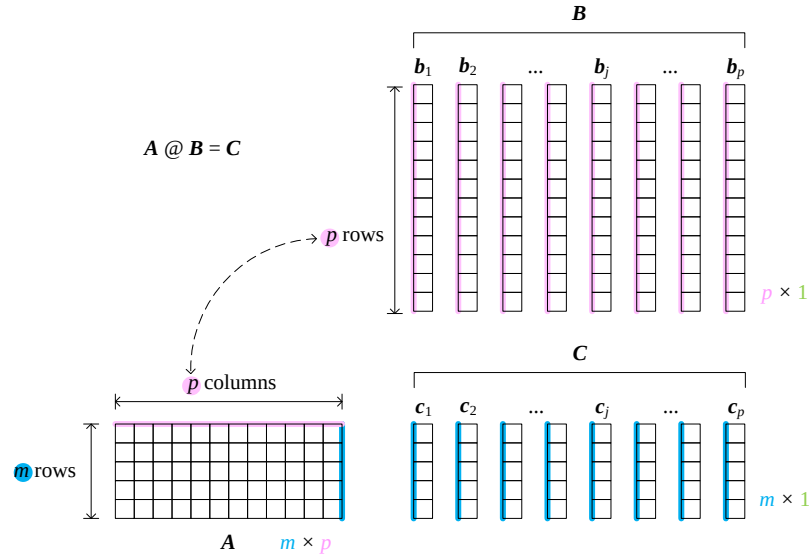
Looking at it the other way around, suppose we have the following set of matrix multiplications:

$$Ab_1 = c_1, \quad Ab_2 = c_2, \quad \dots \quad Ab_p = c_p$$

Among these, the column vectors $b_1, b_2, \dots, b_p$ all have the same shape. The above p equations can be combined to give:

$$A \begin{bmatrix} b_1 & b_2 & \dots & b_p \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & \dots & c_p \end{bmatrix}$$

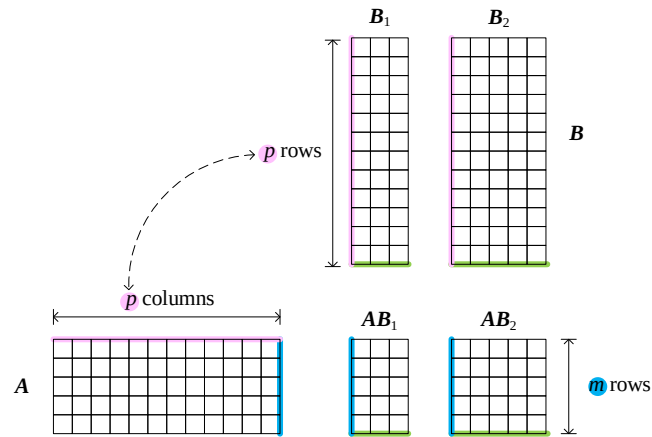
⚠ Please pay special attention to this perspective – this expansion technique appears often in the projection operations covered later in this book.

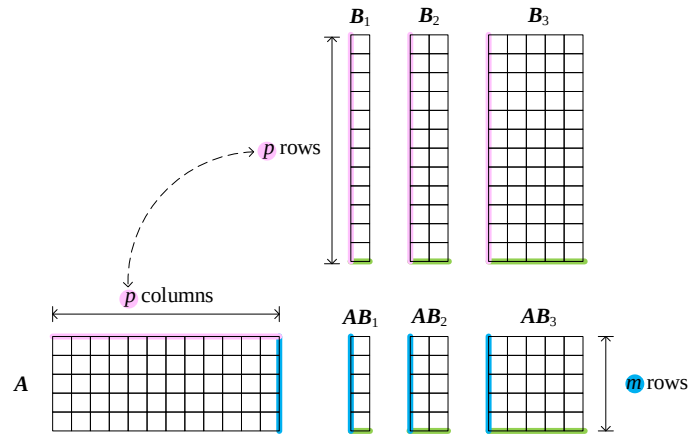
Figure 3. The Matrix Product AB . B Sliced into Columns

Similarly, after first slicing B once from left to right, AB can be expanded and written as:

$$AB = A[B_1 \ B_2] = [AB_1 \ AB_2]$$

Figure 4 shows a diagram of the operation above. Figure 5 tells us that slicing matrix B into several pieces from left to right ultimately doesn't affect the result of the matrix multiplication.

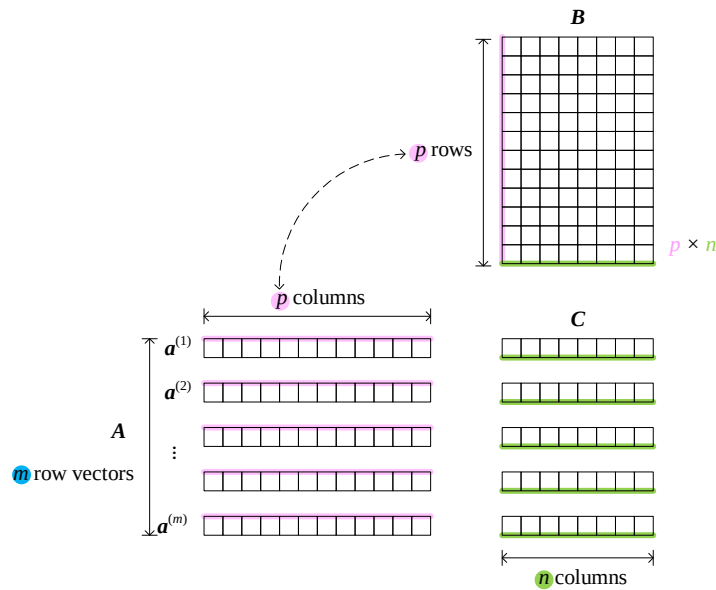
Figure 4. The Matrix Product AB , Slicing B Once from Left to Right

Figure 5. The Matrix Product AB , Slicing B Several Times from Left to Right

The Fourth Perspective on the Matrix Multiplication AB : A Sliced into a Set of Row Vectors

For the matrix multiplication $AB = C$, slicing the left-hand matrix A into rows — as shown in Figure 6, every row of the matrix-multiplication result C is a linear combination of B 's row vectors.

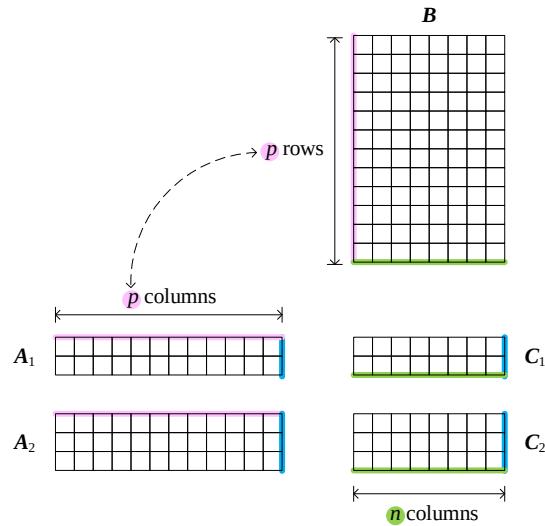
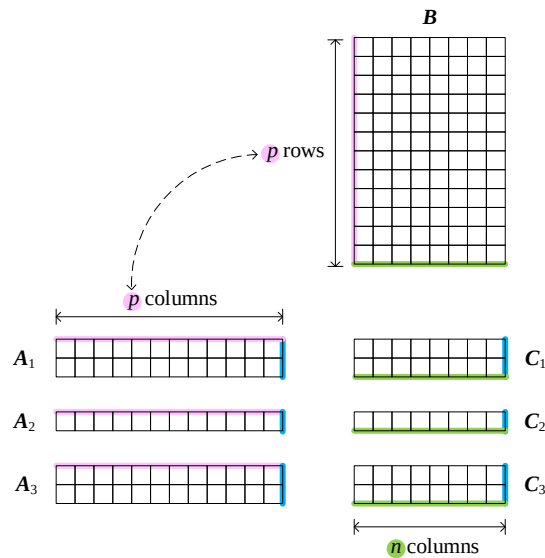
Also, try looking at the operation above in reverse, from the “composition” angle.

Figure 6. The Matrix Product AB , A Sliced into Rows

Slicing A once from top to bottom, the product AB comes out to:

$$AB = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} B = \begin{bmatrix} A_1 B \\ A_2 B \end{bmatrix}$$

Figure 7 shows a diagram of the operation above. Figure 8 tells us that slicing matrix A into several pieces from top to bottom ultimately doesn't affect the result of the matrix multiplication.

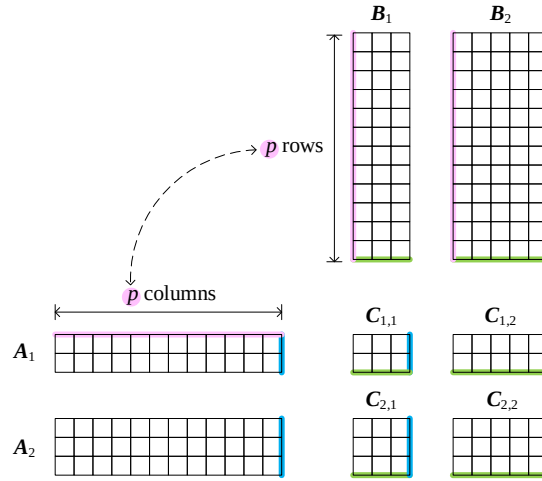
Figure 7. The Matrix Product $\mathbf{AB}$, Slicing $\mathbf{A}$ Once from Top to BottomFigure 8. The Matrix Product $\mathbf{AB}$, Slicing $\mathbf{A}$ Several Times from Top to Bottom

A Sliced Top to Bottom, B Sliced Left to Right

As shown in Figure 9, $\mathbf{A}$ (block-partitioned top-to-bottom) times $\mathbf{B}$ (block-partitioned left-to-right) – the product $\mathbf{AB}$ expands to:

$$\mathbf{AB} = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{bmatrix} \begin{bmatrix} \mathbf{B}_1 & \mathbf{B}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1\mathbf{B}_1 & \mathbf{A}_1\mathbf{B}_2 \\ \mathbf{A}_2\mathbf{B}_1 & \mathbf{A}_2\mathbf{B}_2 \end{bmatrix}$$

We can think of $\mathbf{A}_1$ and $\mathbf{A}_2$ as two elements of matrix $\mathbf{A}$, and $\mathbf{B}_1$ and $\mathbf{B}_2$ as two elements of matrix $\mathbf{B}$. This perspective is similar to the first perspective on matrix multiplication.

Figure 9. The Matrix Product $\mathbf{AB}$. $\mathbf{A}$ Block-Partitioned Top to Bottom, $\mathbf{B}$ Block-Partitioned Left to Right

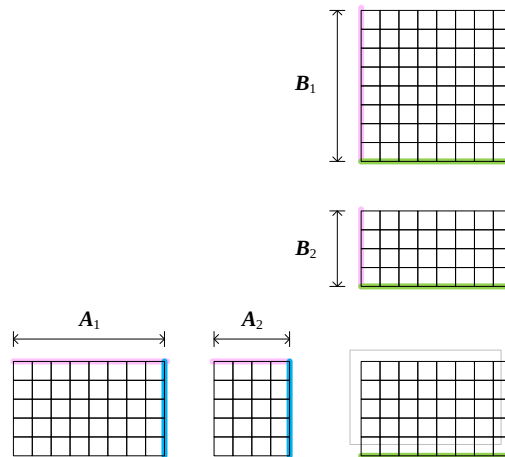
A Sliced Left to Right, B Sliced Top to Bottom

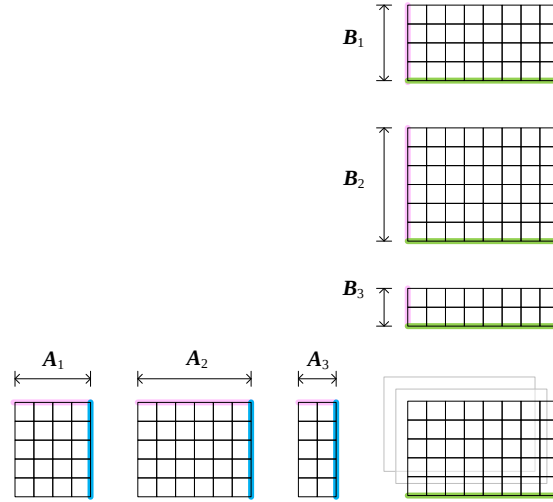
$\mathbf{A}$ (block-partitioned left-to-right) times $\mathbf{B}$ (block-partitioned top-to-bottom) – the product $\mathbf{AB}$ expands to:

$$\mathbf{AB} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 \end{bmatrix} \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{B}_2 \end{bmatrix} = \mathbf{A}_1\mathbf{B}_1 + \mathbf{A}_2\mathbf{B}_2$$

As shown in Figure 10, $\mathbf{A}_1$'s number of columns equals $\mathbf{B}_1$'s number of rows, and $\mathbf{A}_2$'s number of columns equals $\mathbf{B}_2$'s number of rows. As shown in Figure 11, we can also slice $\mathbf{A}$ and $\mathbf{B}$ into even finer pieces.

This is similar to the second perspective on matrix multiplication discussed earlier.

Figure 10. The Matrix Product $\mathbf{AB}$. $\mathbf{A}$ Block-Partitioned Top to Bottom, $\mathbf{B}$ Block-Partitioned Left to Right

Figure 11. The Matrix Product AB . A Sliced into Several Pieces Top to Bottom, B Sliced into Several Pieces Left to Right

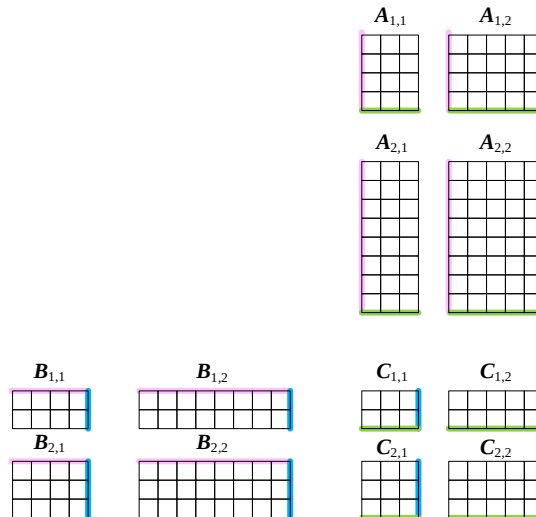
Both A and B Get “Chopped Into Four Pieces”

A and B are both block-partitioned both vertically and horizontally; the product AB comes out to:

$$AB = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{bmatrix} \begin{bmatrix} B_{1,1} & B_{1,2} \\ B_{2,1} & B_{2,2} \end{bmatrix} = \begin{bmatrix} A_{1,1}B_{1,1} + A_{1,2}B_{2,1} & A_{1,1}B_{1,2} + A_{1,2}B_{2,2} \\ A_{2,1}B_{1,1} + A_{2,2}B_{2,1} & A_{2,1}B_{1,2} + A_{2,2}B_{2,2} \end{bmatrix}$$

As shown in Figure 12, $A_{1,1}$, $A_{1,2}$, $A_{2,1}$, $A_{2,2}$'s columns equal, respectively, the number of rows of $B_{1,1}$, $B_{2,1}$, $B_{1,2}$, $B_{2,2}$.

Figure 12 – the block matrix multiplication shown here is equivalent to multiplying two 2×2 matrices; the result C is also 2×2 . This is likewise equivalent to the first perspective on matrix multiplication.

Figure 12. A and B Are Both Block-Partitioned Vertically and Horizontally

This concludes the topic of matrix multiplication in this book; however, the applications of matrix multiplication are only just beginning.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please study the Einstein summation convention on your own.

<https://numpy.org/doc/2.2/reference/generated/numpy.einsum.html>